

0 2 . 1

Table 2 contains information on the digestion of two biological molecules.

Complete **Table 2**.

[4 marks]

Table 2

Biological molecule	Enzyme	Name of bond hydrolysed	Product of digestion
Starch			Maltose
Dipeptide	Dipeptidase		

A student investigated starch digestion by mixing starch with a solution of the enzyme used to digest starch.

The student did a biochemical test for **protein** when starch digestion was completed.

0 2 . 2

Describe a biochemical test to show the presence of protein.

[2 marks]

Question 2 continues on the next page

Turn over ►



0 2 . 3 The student's test for protein was positive.

Explain why.

[2 marks]

0 2 . 4 An enzyme's turnover number (k_{cat}) is the number of substrate molecules converted into product molecules by one enzyme molecule in 1 second. It is determined using this equation.

$$k_{\text{cat}} = \frac{\text{Maximum rate of enzyme-controlled reaction} / \mu\text{mol dm}^{-3} \text{ s}^{-1}}{\text{Enzyme concentration} / \mu\text{mol dm}^{-3}}$$

A scientist investigated the action of a protease enzyme. The scientist prepared a reaction mixture with a protease concentration of $0.0118 \mu\text{mol dm}^{-3}$. The k_{cat} for the protease is 110 substrate molecules per second.

Use this information and the formula to calculate the maximum rate of the protease-controlled reaction.

Give your answer to **3** significant figures.

Show your working.

[2 marks]

Answer _____ $\mu\text{mol dm}^{-3} \text{ s}^{-1}$

10

